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# Pre-IB Mathematics: Calculus from First Principles

The fixed methods for differentiation, integration, indices, surds and logarithms, each stated as a rule first and worked line by line.

**LEVEL** Year 3 and Year 4, pre-IB track · Chapter 1 (Indices) to Chapter 25 (Introduction to Calculus)



## How to use this guide

Each section gives one method as numbered steps, a worked example in which every line carries its reason (the **Why** column, as in our annotated solution manuals), and a **Watch** list of the slips the method prevents. Every method is the **Hit** step of the SHARP routine:

| <b>S</b> See                                                 | <b>H</b> Hit                                             | <b>A</b> Apply                                      | <b>R</b> Refine                               | <b>P</b> Practise                                |
|--------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| Read the command word: is it first principles, or the rules? | Write the rule or definition at the top of your working. | One law or one operation per line, nothing skipped. | Check against the Watch list for that method. | Rework the problem from memory a few days later. |

Sections 1 to 4 are the Year 4 calculus chapters. Sections 5 and 6 are the Year 3 and Year 4 algebra that calculus rests on; revise them first if index laws or logarithms still slow you down.

## 1 Differentiation from First Principles

Year 4 · use when the question says “from first principles”

A first-principles question asks for the derivative from the limit definition, not from the rules. The method never changes, so learn it as a fixed routine.

### Steps

- 1. State the limit definition:**  $f'(x)$  is the limit, as  $h$  tends to 0, of  $[f(x + h) - f(x)] \div h$ .
- Substitute  $x + h$  for every  $x$  in the function.
- Expand and simplify the numerator.
- Cancel the  $h$  in the denominator. The whole method hinges on this move.
- Only then let  $h$  tend to 0.

### WORKED EXAMPLE

Differentiate  $f(x) = x^2$  from first principles. [3]

| Working                                                                               | Why                                       |
|---------------------------------------------------------------------------------------|-------------------------------------------|
| $f'(x) = \text{limit as } h \text{ tends to } 0 \text{ of } [f(x + h) - f(x)] \div h$ | State the definition first.               |
| $f(x + h) = (x + h)^2 = x^2 + 2xh + h^2$                                              | Substitute $x + h$ for $x$ , then expand. |
| $[f(x + h) - f(x)] \div h = (2xh + h^2) \div h$                                       | The $x^2$ terms cancel.                   |
| $= 2x + h$                                                                            | Cancel the $h$ before taking the limit.   |
| As $h$ tends to 0, $f'(x) = 2x$                                                       | Only now let $h$ tend to 0.               |

### Watch: errors this method prevents

- Substituting  $h = 0$  before cancelling, which gives  $0/0$ : undefined, and no marks.
- Writing  $f'(x) = 2x$  by the power rule when the question says “from first principles”.
- Replacing only some of the  $x$  terms with  $x + h$ .

## 2 Differentiation by Rules

Year 4 · use when the question says “differentiate” or “find  $dy/dx$ ”

Once first principles is understood, speed comes from the power rule, applied cleanly one term at a time.

### Steps

1. Rewrite roots and reciprocals as powers first:  $1/x$  becomes  $x^{-1}$ , and the square root of  $x$  becomes  $x^{1/2}$ .
2. **State the rule** at the top of your working: the derivative of  $x^n$  is  $nx^{n-1}$ .
3. Differentiate term by term: bring the power down as a multiplier, then reduce the power by one.
4. Keep every coefficient explicit. A constant term differentiates to 0.

### WORKED EXAMPLE

Find  $dy/dx$  when  $y = 2x^4 - 5x^2 + 3/x - 7$ .

| Working                         | Why                                              |
|---------------------------------|--------------------------------------------------|
| $y = 2x^4 - 5x^2 + 3x^{-1} - 7$ | Rewrite $3/x$ as a power.                        |
| $dy/dx = 8x^3 - 10x - 3x^{-2}$  | Power rule on each term; the constant 7 gives 0. |
| $dy/dx = 8x^3 - 10x - 3/x^2$    | Return the negative power to a fraction.         |

### Watch: errors this method prevents

- Reducing a negative power the wrong way:  $x^{-1}$  becomes  $-x^{-2}$ , not  $x^0$ .
- Differentiating only the denominator of  $3/x$ , which gives 3 instead of  $-3/x^2$ .
- Dropping the coefficient when the power comes down.

## 3 Tangents and Stationary Points

Year 4 · gradient at a point, equation of a tangent, maximum and minimum points

Gradient questions reward a stated procedure. Give each step its reason, and the method marks stay even if the final arithmetic slips.

### Steps

1. **Tangent:** find  $dy/dx$ , evaluate it at the given  $x$  for the gradient  $m$ , find  $y$  at that point, then use  $y - y_1 = m(x - x_1)$ .
2. **Stationary points:** set  $dy/dx = 0$ , solve for  $x$ , and substitute back for  $y$ .
3. **Classify** each point with the second derivative (negative: maximum; positive: minimum) or a sign check either side. If the second derivative is 0, use the sign check.

### WORKED EXAMPLE

For the curve  $y = x^3 - 3x^2 + 4$ , find the tangent at  $x = 1$  and the stationary points.

| Working                                                                                                                                                       | Why                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| $dy/dx = 3x^2 - 6x$                                                                                                                                           | Power rule, term by term.      |
| At $x = 1$ : $m = 3 - 6 = -3$ ; $y = 1 - 3 + 4 = 2$                                                                                                           | Gradient and point of contact. |
| $y - 2 = -3(x - 1)$ , so $y = -3x + 5$                                                                                                                        | $y - y_1 = m(x - x_1)$ .       |
| $3x^2 - 6x = 0$ , so $3x(x - 2) = 0$ , so $x = 0$ or $x = 2$                                                                                                  | Stationary: $dy/dx = 0$ .      |
| Points $(0, 4)$ and $(2, 0)$                                                                                                                                  | Substitute back into $y$ .     |
| $d^2y/dx^2 = 6x - 6$ : at $x = 0$ it is $-6$ , so <b><math>(0, 4)</math> is a maximum</b> ; at $x = 2$ it is $6$ , so <b><math>(2, 0)</math> is a minimum</b> | Second-derivative test.        |

### Watch: errors this method prevents

- Using the value of  $y$ , instead of  $dy/dx$ , as the gradient.
- Finding  $x$  at a stationary point and forgetting  $y$ .
- Naming a maximum or minimum without a test.

## 4 Integration and Areas

Year 4 · indefinite integrals, finding the constant, definite integrals and areas under curves

Integration is differentiation in reverse, with two disciplines that never change: carry the  $+ C$ , and substitute the limits in the right order.

### Steps

1. Decide: **indefinite** (no limits) or **definite** (limits given)?
2. **Reverse the power rule**: add one to the power, then divide by the new power.
3. Indefinite: add  $+ C$ , and find  $C$  from a known point on the curve.
4. Definite: substitute the upper limit, substitute the lower limit, and subtract, in that order. For the area under a curve, integrate between the limits.

### WORKED EXAMPLE

Find  $y$  when  $dy/dx = 6x^2 - 4x + 1$  and the curve passes through  $(1, 5)$ . Then find the area under  $y = x^2 + 1$  from  $x = 0$  to  $x = 3$ .

| Working                                                                                      | Why                                                       |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| $y = \int (6x^2 - 4x + 1) dx = 2x^3 - 2x^2 + x + C$                                          | Add one to each power, divide by the new power, add $C$ . |
| The curve passes through $(1, 5)$ : $5 = 2 - 2 + 1 + C$ , so $C = 4$                         | A known point fixes $C$ .                                 |
| $y = 2x^3 - 2x^2 + x + 4$                                                                    | State the function in full.                               |
| Area under $y = x^2 + 1$ from $x = 0$ to $x = 3$ : $\int_0^3 (x^2 + 1) dx = [x^3/3 + x]_0^3$ | Definite integral between the limits.                     |
| $= (9 + 3) - (0 + 0) = 12$ square units                                                      | Upper limit minus lower limit.                            |

### Watch: errors this method prevents

- Dropping the  $+ C$  on an indefinite integral.
- Substituting only one limit, or subtracting in the wrong order.
- Multiplying by the power, which differentiates instead of integrating.

## 5 Indices and Surds

Year 3 · the laws of indices, simplifying surds, rationalising the denominator

Indices and surds sit under everything that follows, so make the laws automatic. Apply one law per line so each step can be checked.

### Steps

1. **Name the law:** product  $a^m \times a^n = a^{m+n}$ ; quotient  $a^m \div a^n = a^{m-n}$ ; power of a power  $(a^m)^n = a^{mn}$ ; negative  $a^{-n} = 1/a^n$ ; fractional  $a^{1/n}$  is the  $n$ th root of  $a$ .
2. Apply one law per line.
3. Simplify a surd by taking out its largest square factor.
4. Rationalise a denominator by multiplying the top and the bottom by the conjugate.

### WORKED EXAMPLE

Simplify  $(2x^3)^2 \times x^{-4}$ . Simplify  $\sqrt{12}$ . Rationalise the denominator of  $6 \div (3 - \sqrt{3})$ .

| Working                                                                       | Why                                                 |
|-------------------------------------------------------------------------------|-----------------------------------------------------|
| $(2x^3)^2 \times x^{-4} = 4x^6 \times x^{-4}$                                 | Power of a power: square both the 2 and the $x^3$ . |
| $= 4x^2$                                                                      | Product law: $6 + (-4) = 2$ .                       |
| $\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3}$                                   | Take out the square factor 4.                       |
| $6 \div (3 - \sqrt{3}) = 6(3 + \sqrt{3}) \div [(3 - \sqrt{3})(3 + \sqrt{3})]$ | Multiply top and bottom by the conjugate.           |
| $= 6(3 + \sqrt{3}) \div (9 - 3) = 3 + \sqrt{3}$                               | $(a - b)(a + b) = a^2 - b^2$ .                      |

### Watch: errors this method prevents

- Adding powers under a power of a power:  $(x^3)^2$  is  $x^6$ , not  $x^5$ .
- Leaving the coefficient unsquared:  $(2x^3)^2$  is  $4x^6$ , not  $2x^6$ .
- Reading a negative index as a negative number:  $2^{-3}$  is  $1/8$ , not  $-8$ .
- Splitting a root over a sum:  $\sqrt{a + b}$  is not  $\sqrt{a} + \sqrt{b}$ .

## 6 Logarithms and Exponentials

Year 4 · the laws of logarithms, change of base, log and exponential equations

A logarithm is an exponent:  $a^x = y$  means  $x = \log_a y$ . Convert between the two forms fluently, then apply the laws one at a time.

### Steps

1. **State the law you use:** product  $\log_a(xy) = \log_a x + \log_a y$ ; quotient  $\log_a(x/y) = \log_a x - \log_a y$ ; power  $\log_a(x^n) = n \log_a x$ ; change of base  $\log_a x = \log_b x \div \log_b a$ .
2. Isolate the logarithm or the exponential before you solve.
3. Convert to the other form and solve.
4. **Check the domain:** a logarithm takes only a positive argument.

### WORKED EXAMPLE

Solve  $\log_2(x + 3) + \log_2(x - 3) = 4$ . Then solve  $5^x = 40$ .

| Working                                                                             | Why                                 |
|-------------------------------------------------------------------------------------|-------------------------------------|
| $\log_2[(x + 3)(x - 3)] = 4$                                                        | Product law.                        |
| $(x + 3)(x - 3) = 2^4 = 16$                                                         | Convert to exponential form.        |
| $x^2 - 9 = 16$ , so $x^2 = 25$ , so $x = 5$ or $x = -5$                             | Solve the quadratic.                |
| $x = -5$ gives $\log_2(-2)$ , which is undefined, so $x = 5$                        | Domain check.                       |
| $5^x = 40$ , so $x \log 5 = \log 40$ , so $x = \log 40 \div \log 5 = 2.29$ (3 s.f.) | Take logs of both sides; power law. |

**Watch: errors this method prevents**

- Splitting a log over a sum:  $\log(x + y)$  is not  $\log x + \log y$ .
- Keeping a solution that makes an argument zero or negative.
- Converting the wrong way:  $\log_2 16 = 4$  because  $2^4 = 16$ .

**Summary: every framework on one page**

| Question type               | Method, in order                                                                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First principles            | State the limit definition; substitute $x + h$ ; expand and simplify; cancel $h$ ; only then let $h$ tend to 0.                                                 |
| Differentiation by rules    | Rewrite roots and reciprocals as powers; state the power rule; bring the power down and reduce it by one, term by term; constants give 0.                       |
| Tangent at a point          | $dy/dx$ ; gradient $m$ at the given $x$ ; the point $(x_1, y_1)$ ; $y - y_1 = m(x - x_1)$ .                                                                     |
| Stationary points           | Set $dy/dx = 0$ ; solve for $x$ ; substitute back for $y$ ; classify by the second derivative or a sign check.                                                  |
| Integration and areas       | Add one to the power, divide by the new power; $+ C$ for indefinite, found from a known point; definite: upper minus lower; area: integrate between the limits. |
| Indices and surds           | Name the law; one law per line; take out square factors; rationalise with the conjugate.                                                                        |
| Logarithms and exponentials | State the law; isolate the log or the exponential; convert form and solve; check every argument is positive.                                                    |

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